

# BRANCH DOCUMENT — IRIS-3

Branch: branch\_20260512\_003051 | Base: main-v1.0

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Decision: NO\_GO | Confidence: 53%

## CHANGE REQUEST

vorrei ingrandire la struttura arrivando a farla pesare 36kg

## CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Mass constraint violation

Subsystem: structure | Req: L1-MAS-001 | Type: constraint\_violation

The total mass of the payload exceeds the platform accommodation envelope by 1 kg, violating the requirement L1-MAS-001.

[MAJOR] ISSUE-002 — Thermal margin reduction

Subsystem: thermal | Req: L1-THM-001 | Type: reanalysis\_required

Increasing the mass of the structure may alter thermal properties and reduce the  $\Delta T$  margin, potentially violating L1-THM-001.

[MINOR] ISSUE-003 — Potential structural integrity impact

Subsystem: mechanical | Req: None | Type: reanalysis\_required

Increasing the mass of the structure may affect its structural integrity and natural frequencies, requiring reanalysis.

## MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-32C Rev.1 §9 — Shock test per launcher SRS ICD

Reason: The change may affect the structural dynamics, necessitating a reevaluation of shock response.

## SUBSYSTEM STATUS — UPDATED

| Subsystem   | Score | Status | Detail                                | Changed    |
|-------------|-------|--------|---------------------------------------|------------|
| optical     | 92%   | OK     | MTF 0.42 > threshold 0.40             | —          |
| thermal     | 41%   | WARN   | Thermal margin reduction              | ■ MODIFIED |
| mechanical  | 80%   | OK     | Potential structural integrity impact | ■ MODIFIED |
| structure   | 56%   | FAIL   | Mass constraint violation             | ■ MODIFIED |
| solar       | 89%   | OK     | Power margin +18%                     | —          |
| startracker | 67%   | WARN   | Stray light analysis pending          | —          |
| dataproc    | 71%   | WARN   | Compression ratio TBC                 | —          |

## AFFECTED REQUIREMENTS

- L1-MAS-001 — verification evidence invalidated, re-analysis required
- L1-THM-001 — verification evidence invalidated, re-analysis required